

Hybrid Ecosystem Simulator

Parameter Manual

Ecosystem Simulation Frontend

April 2026

1 General parameters

Parameter	Range/Default	Description
Number of Species (NSPP)	1–10 / 5	Number of plant species with unique nutrient niches
Grid Height/Width (H,W)	10–500 / 100	Spatial simulation grid dimensions
Max Agents	10k–500k / 150k	Maximum number of individual plants simulated
Steps (NSTEPS)	100–5000 / 1500	Total simulation timesteps
Seed/NSeeds	0–9999 / 35, 1–20 / 1	Random seed + ensemble runs for averaging
Scalar interval	1–500 / 20	Record biomass/agents/elements every N steps
Snapshot interval	1–500 / 50	Save full grid maps every N steps

2 Biological rates

Parameter	Range/Default	Description
Growth Rate	0.01–1.0 / 0.45	Biomass growth efficiency from C uptake
Respiration Rate	0.001–0.1 / 0.015	Lost of biomass/step
Turnover Rate	0.001–0.1 / 0.03	Biomass-to-soil recycling rate
Mineralization Rate	0.01–0.2 / 0.05	Organic-to-available nutrient conversion
Seed Cost	0.001–0.1 / 0.02	Biomass cost to produce seeds
Seed Mass	0.001–0.1 / 0.02	Initial mass of new seeds
K Biomass	0.5–10.0 / 2.5	Carrying capacity parameter
Sigma Threshold	0.1–5.0 / 3.0	Niche fitness sensitivity (Higher less penalty for deviations from the center of biogeochemical niche)

3 Soil parameters

Parameter	Range/Default	Description
Soil Pool Mean	0.5–3.0 / 1.5	Inorganic nutrient baseline
Soil Pool Std	0.01–0.5 / 0.1	Initial spatial variation
Soil Ratio Noise	0.0–0.2 / 0.05	Randomness in N/P/K/O ratios
Soil Input Rate	0.1–2.0 / 0.5	External nutrient inflow
Input Drift Scale	0.0–0.3 / 0.08	Environmental fluctuation noise
Soil Base Ratio (N,P,K,O)	0–1.0 / 0.35,0.10,0.35,0.10	Soil elementome
Soil Availability (N,P,K,O)	0–1.0 / 0.4,0.1,0.1,0.3	Percentage of available nutrients per timestep

4 Stochastic disturbances

Parameter	Range/Default	Description
Catastrophe Interval	50–500 / 200	Steps between mass mortality events
Catastrophe Mor- tality	0.0–0.9 / 0.4	Fraction of agents killed per event
Disturbance Prob.	0.0–0.05 / 0.01	Probability/step of soil depletion patch
Disturbance Strength	0.0–1.0 / 0.7	Fraction of soil nutrients removed
Demographic Noise	0.0–0.01 / 0.003	Std of random mass perturbation/step

5 Species traits

- **Niche Centers:** Center of biogeochemical niche (Carbon, Nitrogen, Potassium, Phosphorus, others. Concentration)
- **Covariance Matrix:** Niche tolerance (Size of biogeochemical niche)